

Methods

LASSO Procedure

To identify which predictor scales most strongly account for variance in smartphone addiction severity, LASSO regression is applied using a Gaussian family specification, treating SPAI total as a continuous outcome. All individual SPAI items are excluded from the predictor set to prevent circularity, as they directly constitute the outcome variable. The remaining predictors include all items from the NMP, SAPS, MPAS, MPATS, and SAS scales alongside demographic variables (age, gender, and grade level).

Formally, LASSO minimizes the penalized loss function as shown in the equation below:

$$\hat{\beta} = \arg \min_{\beta} \sum_{i=1}^n (y_i - x_i^T \beta)^2 + \lambda \sum_j |\beta_j|$$

where the first term is the standard residual sum of squares and the second term is the L1 penalty applied across all p predictors. The tuning parameter λ controls the severity of shrinkage as λ increases, more coefficients are driven to exactly zero and removed from the model entirely (Tibshirani, 1996).

The penalty parameter λ is selected via 10-fold cross-validation, evaluating both λ -min and λ -1se. The former is essentially the best predictive fit on average. The latter is almost as good but it simple as it retains the fewest predictors necessary for a competitive level of explained variance. The full sample is partitioned into an 80% training set and a 20% held-out test set, with all model fitting occurring exclusively on the training data. Model performance is then evaluated on the held-out test set using R^2 and RMSE.